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To: Dr. Terry Xu

From: Mikey Wiesecke MEGR 3152 Section L09

Identification of a Unknown Sample

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Abstract

The goal of this experiment was to identify the identity of a sample with a series of tests. These tests included the tensile test, metallography examination, hardness test, magnet test, and a density test. Through these tests it was discovered that the unknown material was not ferrous and had a density of 2.379. It was then concluded that the material had to be some type of aluminum and through the metallography test sample was confirmed to be an aluminum alloy. Once a tensile test and a Rockwell hardness test was done it was confirmed that the sample had a yield strength of 372 MPa and a HRB of 81.22. Matching these characteristics, it was discovered that the best matching sample was 2014-T6 Aluminum.

Objective

The objective of this experiment was to identify an unknown sample using multiple tests. These tests included Tensile test, Metallurgy examination, Rockwell Hardness Test, and a density test. This experiment shows how these tests that were learned in the lab could be used to identify an unknown sample showing real life practicality.

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Introduction

In this the experiment “Identification of an Unknown Sample” a sample whose material composition was unknown was tested and analyzed in order to find composition of the sample. This was done using the tensile test, Magnet test, a Metallurgy evaluation, the Rockwell hardness test, and a density test. The tensile test involved the use of an Instron 5582 testing machine which was equipped with a 6800 series retrofits. The sample was pulled apart by a tensile test with increasing force while measuring the distance between two set points on the sample. Using the data from this machine a stress versus strain graph was made. The stress strain graph was analyzed to find the yield strength. young’s modulus using Hooks law and the .002 offset method. The magnet test was done to determine whether the sample was a ferrous element with ferrous element being attracted to the magnet. Using the magnet test and tensile test it was decided whether the general material was steel, aluminum, or brass. However, the specific composition of the material was still a mystery and more tests needed to be conducted. The Metallography examination was done by polishing the sample using multiple steps of sandpaper and polishers to prepare the sample. The prepared sample was then examined under an Olympus BX51 Microscope to reveal the sample’s microstructure. The microstructure told the different phases that were present as well as grain size, grain shape, and participants. The microstructure was crucial in determining the specific composition. Once a good guess was determined using the microstructure and stress properties, the Rockwell Hardness test was used to verify the results. The Rockwell hardness test was done by placing the sample on a Wilson 2000 Rockwell Hardness testing machine and running test for 5 trials. If the material was ferrous then the HRG and HRC scale were used with the HRC testing needing a diamond tip test. If the material was nonferrous and likely soft, then the HRB scale was used which could be done with the **standard 1/16th in steel ball tip**. This was best for an unknown sample because the hardness is a direct indicator of carbon percentage in steel. If the sample had a high hardness, then it was likely that the steel was some sort of high carbon steel that had not been heat treated. For non-steel alloys hardness was not as informative but could be used to determine whether the alloying material made the base material stronger or weaker. If it was known that the sample was an aluminum alloy it could be determined if the sample was alloyed with a material stronger than aluminum based on the hardness number. Knowing how to determine the composition of an unknown sample could be incredibly important and useful. For example, when a plane crashes it’s very difficult to identify the cause of scraps but if these scraps could be identified and analyzed then it would be possible to discover the source of failure. Being able to determine the material of an unknown sample could determine where a piece of scrap might have come from.

Equipment and Procedure

The first test that was done was the magnet test which required a magnet. The test was to simply bring the sample close to the magnet and observe for any attraction. The next test was the tensile test which needs the Instron 5582 testing machine with 6800 series retrofits and the unknown sample. Two dots were placed on the middle portion of the sample about 1.8in away from each other. The radius of the sample was also measured and was entered into the machine

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before the test was started. The 5582 Testing machine pull rate was set to 5.08mm per minute and was set to pull to failure. After securing the sample into the testing machine the sample was pulled until failure occurred and recorded the change in distance between the two points as well as the change in force. The tensile test was complete, and the sample could undergo an metallography examination. The unknown sample was prepped by sanding with 320, 400, and 600 grit sandpaper. The magnet test verified that the unknown sample was nonferrous and likely a soft material which meant the 240-grit sandpaper was not needed for this specific sample. After the sample was sanded down to ensure that the old scratch marks were replaced with new perpendicular scratch marks and the sample was moved to the polishing room. For hard samples a round of polishing on the 800-grit polishing wheel was required while running the wheel at 200 rpm. However, because the sample was hypothetically a soft material the polishing wheel was run at 100 rpm and only the 1200 grit wheel was needed. Once polished using the 1200 grit polishing wheel, the sample was cleaned and dried with compressed air to be inspected. Once adequate polishing on the 1200 grit polishing wheel was done the sample was polished again using the micro cloth polishing wheel with $0.3\ \mu\text{m}\ \text{Al}_2\text{O}_3$ suspension. Once washed and dried the sample was then inspected under an Olympus BX51 Microscope to search for any pre etching features which would be an indicator of steel. The final step for prepping the sample for the Metallography sample was to use an etching agent to expose the subsurface microstructure. The unknown sample was etched for a total of 15s. The sample was then again inspected under an Olympus BX51 Microscope and pictures of the microstructure were taken at 100x, 50x, and 20x magnification. The pictures were saved and shared to be analyzed later. The next test required the use of a Wilson 2000 Rockwell Hardness testing machine which was displayed in figure 1 below along with the Olympic BX51 Microscope used in previous experiment.

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Figure 1. Picture of the Wilson 2000 Rockwell Hardness Testing machine(left) and a Olympus BX51 Microscope(right)

The final test which could be done at any time or order was the density measurement test. The density measurement required two variables, weight, and volume. Volume was found by filling a beaker with water to any level. This level was set at the initial volume, the sample was then placed in the beaker and the new water level was recorded as the final volume. Volume was equal to the difference between the final volume and the initial volume. To get the weight of the sample a Mettler -PM 1200 scale was used which had a reliability of 0.0001g, reproducibility of 0.001g and a linearity of $\pm 0.002g$. The weight of the sample was measured after the scale is zeroed and the box surrounding the scale was closed. The unknown sample density was found by dividing the weight of the sample the samples volume. In order find the uncertainty of the density measurement equation 1 could be used.

$$\delta D = \sqrt{\left(\frac{1}{V} \delta w\right)^2 + \left(\frac{-w}{V^2} \delta V\right)^2} \quad \text{Eq.1}$$

δD was the uncertainty of the density and δw was the reproducibility of the scale which was 0.001g. The δV was the uncertainty of the volume measurement. Therefore, the final density was presented as $D \pm \delta D$.

Result and Discussion

The magnet test is a test to determine the presence of ferrous elements in the sample. Following this test, no ferrous elements are present. This is because the magnet is not attracted to the sample. Another thing that can be noted is the yellowish silver color of the sample which is a key characteristic of aluminum. Steel is much darker than aluminum and is attracted to magnets because of the presence of ferrous elements in steel. The tensile test reports the yield stress and young's modulus on whether the material is ductile or brittle. Figure 2 is a stress versus strain chart of the unknown sample.

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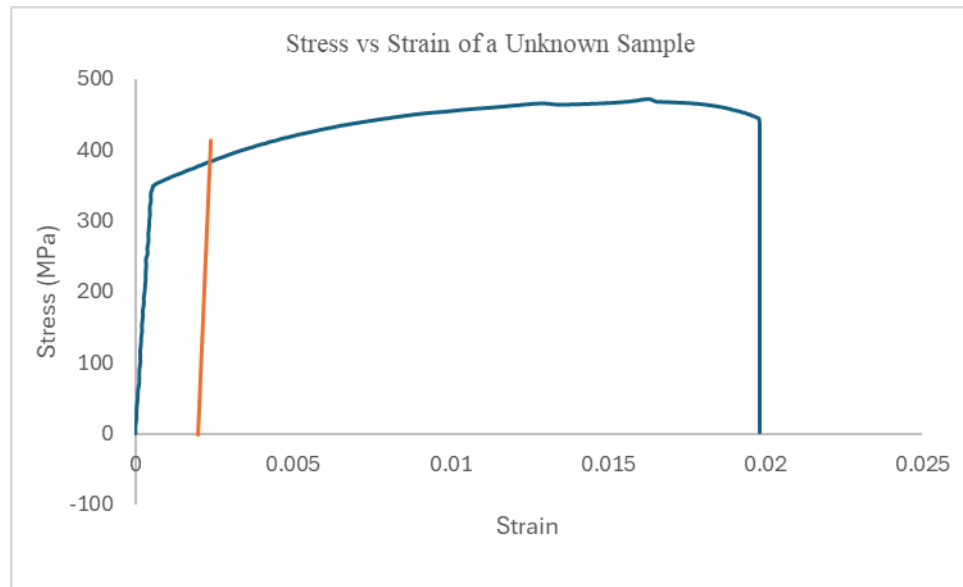


Figure 2. Stress versus Strain of a Unknown Sample

The Graph shows the material to be much more brittle than normal aluminum with an increased yield stress. This suggest the presence of a stronger material diluted into the aluminum. The yield stress is 372.97 MPa and with a young modulus of 105.1MPa. In contrast the T6 Aluminum tested in a previous lab has a yield stress of 275.79 MPa and an elastic modulus of 68.9 MPa. This is evident from the failure mechanism and cross section of the fractured sample. The sample does not undergo very much necking which is verified by the graph. Figure 3 shows a picture of the fracture sample.

The main cause of failure is two cracks forming during the test which propagates and ultimately causes the fracture. The surface itself does not show a cup and cone fracture but rather a rough almost brittle fracture. However, the fracture itself is not a clean cut and cannot be considered a brittle fracture, the material undergoes a ductile fracture. Now that it is known that the unknown material is an alloyed aluminum and has been strengthened the only other test is to confirm these facts with a metallography examination. Figure 4 shows the microstructure of the unknown samples at different magnification.

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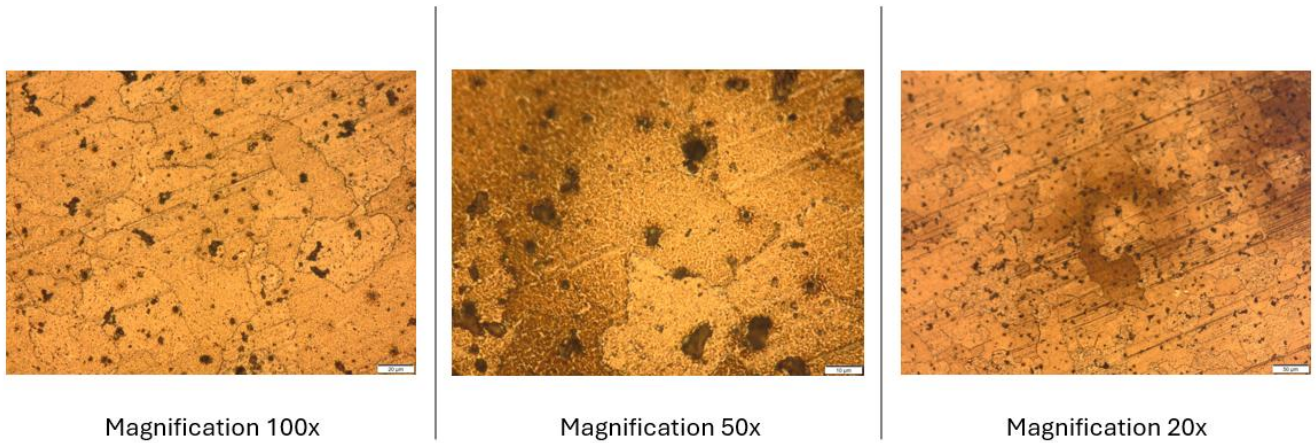


Figure 4. Microstructure of Unknown sample at different magnification

The Microstructure shows yellow phases and darker phases, this darker phase is the Aluminum while the yellow phases is the unknown alloying element. There is also the presence of precipitance in the microcenter which suggests that the material is alloyed. The precipitants could be magnesium, nickel or silicon added in when alloying the aluminum. Based on the yellow color and the fact that the unknown alloy is stronger than aluminum it can be predicted that this material is some type of copper alloyed aluminum. The last test to get the final numbers used to determine the unknown material is the Hardness test. The Rockwell Hardness test results are displayed in the table below, table 1.

Table 1 . Rockwell Hardness test Results for Unknown Sample

Rockwell Hardness Test	
HRB	
Trial 1	80.8
Trial 2	82.1
Trial 3	80.8
Trial 4	80.3
Trial 5	82.1
Average	81.22
STD	0.8288546

The sample has an average hardness of 81.22 on the HRB scale with a standard deviation of 0.829. Matching all available information there is a strong possibility that the Aluminum could be 2014-T6 Aluminum. 2014-T6 Aluminum has a hardness of 82 HRB and a yield strength of 365 MPa which is almost identical to our sample. The hardness of the sample is 81.22 with a deviation of .829 which added together is HRB 82.049. This is almost identical to the HRB reported for 2014-t6 Aluminum. In addition to the hardness, the yield strength of 2014-T6 Aluminum is almost the same as the unknown sample. The unknown sample is 365MPa and the

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2014-T6 Aluminum has a yield strength of 372MPa which is a estimated value. To make sure of the material the results from the density test can be used. The volume of the unknown sample is around 8.6614mm cubed and has a weight of 7.137g. This means the density is 2.379 g/cm³ slightly lower than the density of aluminum however because the measurements were done by eye there is the possibility of error.

Conclusions

The main objective of this experiment was to discover through different tests the identity of a unknown sample. Though these it was discovered that the material is nonferrous. Based off the color and density it was concluded that the material was aluminum. It was discovered that the aluminum was not the baseline aluminum as it was significantly stronger and harder than the baseline aluminum. This was done through the tensile test and the Rockwell Hardness test which resulted in a yield strength of 372 MPa and hardness of HRB 82. The conclusion was that the material was some type of alloyed aluminum and was hardened primarily by copper. A metallography examination was done to confirm the presence of alloying elements in the aluminum. From there the numbers were matched between different alloyed aluminums and it was determined that 2014-T6 Aluminum was the best match and was most probably the identification of the unknown material. Of course, there was a possible error in the density test because the water level was measured by eye which might explain why the density did not match up exactly. There could have also been an error in the yield strength because the .002 offset method was only a estimation of the yield strength. The Hardness test was by far the most accurate result matching exactly with the listed specs of the 2014 T6 aluminum. Being able to identify an unknown sample allows for the ability to track items or parts. In the case of a failure, it is key to know which component failed and when left with scraps it can be hard to figure out where the failed component came from. Knowing how to identify a material can help figure out which component failed and how it failed.

References

1. The UNC Charlotte William States Lee College of Engineering
Department of Mechanical Engineering and Engineering Science. 2024. "Lab Instruction 6_Unknown Samples_Spring 2024". Last modified Spring 2024.
<https://uncc.instructure.com/courses/218553/pages/topic-6-identification-of-an-unknown-sample>

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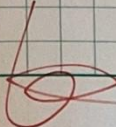
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Appendices

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- Test 5 - Density [remove off arm previous test] + do this before etching o To find the density the weight and volume must be found <u>Volume</u> ↳ Prepare a glass cylinder and fill it with water up to a certain point, mark this point as $V_{initial}$ ↳ Gently drop sample into cylinder and mark the new water level as V_{final} $V = V_{final} - V_{initial}$ ↳ do the test at least 3 times with the uncertainty being the standard deviation of the volume. <u>Weight</u> - o Use a Mettler-PM 1200 scale in the microscope lab, ↳ The scale has a capacity of 1200g or 1.2kg and a readability of 0.0001g, reproducibility of 0.0001 0.001g, linearity of $\pm 0.002g$ ↳ select the unit as it matches the water cylinder unit ↳ Set scale to zero ↳ lift lid and gently put sample on scale ↳ close lid and take weight reading o Density is given by $\rho = \frac{w}{V}$ w: weight V: Volume ρ: density o uncertainty is determined by $SD = \sqrt{C \pm SD_w)^2 + (C \pm \frac{w}{V} SD_V)^2}$ SD is reproducibility of scale o Report density as $D_{sample} \pm SD$ E		
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